

Project Abstract

Smart India Hackathon 2026 | Problem Statement SIH26153 | CYBERMIND

Abstract

Modern intrusion-detection systems primarily describe traffic that has already occurred, giving defenders limited support for anticipating attack progression or comparing possible responses. CYBERMIND addresses SIH26153 through an offline graph-temporal world model for short-horizon cyber-risk forecasting. The platform transforms packet captures or network-flow records into rolling host-communication graphs, where hosts form nodes and communications form feature-rich edges. A GATv2 encoder learns the spatial structure of each graph, a temporal Transformer models changes across observed windows, and a learned latent dynamics module rolls the network state forward. Multi-head outputs estimate future infiltration risk, coarse attack stage, and predictive dispersion.

The forecasting engine is integrated with an analyst workspace rather than presented as an isolated model. The Command Centre connects observed topology, flow evidence, and a future risk timeline. Parallel Futures compares the same model's baseline forecast against controlled in-memory graph modifications representing actions such as rate limiting, port blocking, or host isolation. A bounded offline assistant answers case-grounded questions, while a local validation engine records real sandbox HTTP responses and SHA-256 hashes beside a frozen forecast. For secure collaboration, incident briefs are encrypted locally with AES-256-GCM and decrypted inside the application using a separately shared key.

The selected checkpoint achieved pooled held-out binary F1 of 0.9849, precision of 0.9970, recall of 0.9730, and a false-positive rate of 0.0056 on the documented capture-grouped CIC-IDS2018 four-observed-window to four-future-window protocol. On that split, it outperformed a feature-matched logistic-regression baseline. These results support short-horizon binary graph-window forecasting under the stated evaluation conditions; they do not establish universal performance or verified per-flow attack labels.

CYBERMIND therefore demonstrates a practical transition from retrospective alerting toward evidence-linked prediction, counterfactual simulation, and informed defensive planning. Its core pipeline runs locally through a Windows desktop application or offline Docker package, keeping operational data and model inference under the analyst's control. Stage estimates, uncertainty indicators, and simulated interventions are clearly presented as decision aids, while consequential actions remain subject to human approval.

Keywords

Predictive cyber defence, graph neural network, temporal Transformer, network world model, attack forecasting, counterfactual simulation, offline security analytics, AES-256-GCM.